Mutation → Behavior Class Map — Implementation Roadmap

> **Purpose**: This document converts the qualitative gene→behavior map (21 genes × 6 axes) into sequenced, copy-pasteable AI prompts. It is a **companion** to the Stroma World Implementation Roadmap — it refines and extends the Boolean network, phenotype mapping, fitness function, and validation layers with the biological nuance from the behavior class map.
> **Dependency**: Complete Phases 0–1 of the Stroma World roadmap first. This roadmap's prompts replace or extend specific components from Phases 1–7.
> **Key additions over the base roadmap**:
> - Multi-axis effect matrix (each gene affects multiple behavioral axes, not just one)
> - SMAD4 asymmetry logic (the most critical biological feature)
> - Go-vs-grow tradeoff enforcement
> - Drug-conditional ABCB1 behavior
> - Clinical trial failure reproduction as validation
> - Confidence-weighted parameterization

Phase A: Effect Matrix Data Structure

Prompt A.1 — Gene Effect Matrix as Code

```
Create a new file: custom_modules/gene_effect_matrix.h


This file encodes the full 21-gene × 6-axis effect matrix as a compile-time data structure.
Every gene in Stroma World affects multiple behavioral axes with varying strength and direction.
This matrix is the SINGLE SOURCE OF TRUTH for how mutations map to phenotype changes.


ENUMS:
```

```
enum class BehaviorAxis {
    GROWTH_DRIVE = 0,      // Proliferative cycling rate
    DEATH_RESISTANCE = 1,  // Evasion of apoptosis and therapy-induced death
    CELL_CYCLE_BRAKING = 2,// Responsiveness to arrest signals (HIGHER = more responsive to brakes)
    MOTILITY_INVASION = 3, // Migration speed, ECM remodeling, mesenchymal phenotype
    PRO_STROMA_SIGNAL = 4, // Tendency to activate stroma and build ECM barrier
    IMMUNE_EVASION = 5,    // Immune suppression (out of scope for Cycle 1, but tracked)
    AXIS_COUNT = 6
};

enum class EffectDirection {
    STRONG_INCREASE = 3,    // ↑↑↑
    MODERATE_INCREASE = 2,  // ↑↑
    MILD_INCREASE = 1,      // ↑
    NEGLIGIBLE = 0,         // ——
    MILD_DECREASE = -1,     // ↓
    MODERATE_DECREASE = -2, // ↓↓
    STRONG_DECREASE = -3,   // ↓↓↓
    CONTEXT_DEPENDENT = 99  // ◆ — requires runtime conditional logic, cannot be a static value
};

enum class Confidence { HIGH, MED, LOW };

enum class CellTypeScope { TUMOR, STROMA, BOTH };

STRUCT:

struct GeneAxisEffect {
    EffectDirection direction;
    Confidence confidence;
    bool is_conditional;          // true if effect depends on another gene's state or substrate level
    const char* condition_desc;   // human-readable: e.g. "Only when SMAD4 is lost"
};
```

```
struct GeneBehaviorProfile {
    GeneIndex gene;
    const char* mutation_type;    // "GoF", "LoF", "Overexpr", "HomDel", "Suppressed", etc.
    CellTypeScope scope;
    GeneAxisEffect effects[6];   // indexed by BehaviorAxis
    const char* critical_caveat;  // most important modeling note for this gene
};

STATIC DATA — populate the full 21-gene matrix exactly as follows:

// Gene 1: KRAS (gain-of-function G12D/G12V/G12R)
{
    GeneIndex::KRAS, "GoF", CellTypeScope::TUMOR,
    {
        {STRONG_INCREASE, HIGH, false, nullptr},       // GD: ↑↑↑ — root driver of proliferation
        {MODERATE_INCREASE, HIGH, false, nullptr},     // DR: ↑↑ — PI3K/AKT survival signaling
        {MODERATE_DECREASE, HIGH, false, nullptr},     // CCB: ↓↓ — overrides growth arrest signals
        {MILD_INCREASE, HIGH, false, nullptr},         // MI: ↑ — RalGEF pathway
        {STRONG_INCREASE, HIGH, false, nullptr},       // PSS: ↑↑↑ — drives TGFβ, SHH, IL-6 secretion
        {MODERATE_INCREASE, HIGH, false, nullptr}      // IET: ↑↑ — GM-CSF, PD-L1
    },
    "Root node. Every axis is downstream of KRAS. Growth drive is most direct."
},

// Gene 2: MYC (amplification / transcriptional upregulation)
{
    GeneIndex::MYC, "Amp/Overexpr", CellTypeScope::TUMOR,
    {
        {STRONG_INCREASE, HIGH, false, nullptr},       // GD: ↑↑↑
        {MILD_INCREASE, MED, true, "Pro-survival only when TP53 is lost; otherwise MYC sensitizes to
apoptosis"},
        {STRONG_DECREASE, HIGH, false, nullptr},       // CCB: ↓↓↓ — overrides checkpoints
```

```
        {NEGLIGIBLE, HIGH, false, nullptr},          // MI: ——
        {MILD_INCREASE, MED, false, nullptr},        // PSS: ↑ — indirect via metabolic hypoxia
        {MILD_INCREASE, MED, false, nullptr}         // IET: ↑
    },
    "Go-vs-grow tension: MYC high + ZEB1 high should be penalized (antagonistic)."
},

// Gene 3: EGFR (overexpression / autocrine activation)
{
    GeneIndex::EGFR, "Overexpr", CellTypeScope::TUMOR,
    {
        {MODERATE_INCREASE, MED, false, nullptr},     // GD: ↑↑ — redundant with KRAS
        {MILD_INCREASE, MED, false, nullptr},        // DR: ↑
        {MILD_DECREASE, MED, false, nullptr},         // CCB: ↓
        {MILD_INCREASE, MED, false, nullptr},        // MI: ↑ — EGFR→RAS→RAC
        {NEGLIGIBLE, MED, false, nullptr},           // PSS: ——
        {NEGLIGIBLE, MED, false, nullptr}            // IET: ——
    },
    "Clinically disappointing target (erlotinib marginal). Functions as modulatable input node for EA."
},

// Gene 4: TP53 (loss-of-function / dominant-negative missense)
{
    GeneIndex::TP53, "LoF", CellTypeScope::TUMOR,
    {
        {MILD_INCREASE, HIGH, false, nullptr},       // GD: ↑
        {STRONG_INCREASE, HIGH, false, nullptr},      // DR: ↑↑↑ — apoptosis disabled
        {STRONG_DECREASE, HIGH, false, nullptr},      // CCB: ↓↓↓ — G1/S and G2/M checkpoints gone
        {MODERATE_INCREASE, MED, false, nullptr},     // MI: ↑↑ — GoF mutants promote invasion via
integrin/RhoA
        {MILD_INCREASE, MED, false, nullptr},        // PSS: ↑ — cooperates with KRAS on TGFβ/SHH
        {MILD_INCREASE, MED, false, nullptr}         // IET: ↑
    },
```

```
    "Treat as LoF for Cycle 1. GoF missense effects (R175H, R273H) are a Cycle 2 refinement."
},

// Gene 5: BCL-XL / BCL2L1 (overexpression)
{
    GeneIndex::BCL_XL, "Overexpr", CellTypeScope::TUMOR,
    {
        {NEGLIGIBLE, HIGH, false, nullptr},          // GD: ——
        {STRONG_INCREASE, HIGH, false, nullptr},     // DR: ↑↑↑ — sequesters BAX/BAK
        {NEGLIGIBLE, HIGH, false, nullptr},          // CCB: ——
        {NEGLIGIBLE, HIGH, false, nullptr},          // MI: ——
        {NEGLIGIBLE, HIGH, false, nullptr},          // PSS: ——
        {NEGLIGIBLE, HIGH, false, nullptr}           // IET: ——
    },
    "Clean single-axis gene. Navitoclax causes thrombocytopenia — fitness should include toxicity cost."
},

// Gene 6: BAX (functionally suppressed by BCL-XL, not mutated)
{
    GeneIndex::BAX, "Suppressed", CellTypeScope::TUMOR,
    {
        {NEGLIGIBLE, HIGH, false, nullptr},          // GD: ——
        {MODERATE_DECREASE, HIGH, true, "This row shows effect of BAX SUPPRESSION: less death = more
resistance. BAX active = pro-death."},
        {NEGLIGIBLE, HIGH, false, nullptr},          // CCB: ——
        {NEGLIGIBLE, HIGH, false, nullptr},          // MI: ——
        {NEGLIGIBLE, HIGH, false, nullptr},          // PSS: ——
        {NEGLIGIBLE, HIGH, false, nullptr}           // IET: ——
    },
    "BAX is the node you WANT active. Apoptosis fires when BAX > BCL-XL. Not a drug target itself."
},

// Gene 7: CDKN2A / p16 (homozygous deletion / promoter methylation)
```

```
{
    GeneIndex::CDKN2A, "HomDel", CellTypeScope::TUMOR,
    {
        {MODERATE_INCREASE, HIGH, false, nullptr},     // GD: ↑↑ — consequence of losing brake
        {NEGLIGIBLE, HIGH, false, nullptr},            // DR: ——
        {STRONG_DECREASE, HIGH, false, nullptr},       // CCB: ↓↓↓ — p16→CDK4/6→RB brake removed
        {NEGLIGIBLE, HIGH, false, nullptr},            // MI: ——
        {NEGLIGIBLE, HIGH, false, nullptr},            // PSS: ——
        {NEGLIGIBLE, HIGH, false, nullptr}             // IET: ——
    },
    "Growth increase is a CONSEQUENCE of losing the brake, not independent mitogenic signaling.
CDKN2A=OFF releases RB, does not directly activate MYC."
},

// Gene 8: SMAD4 / DPC4 (homozygous deletion / inactivating mutation)
{
    GeneIndex::SMAD4, "LoF", CellTypeScope::TUMOR,
    {
        {MILD_INCREASE, HIGH, false, nullptr},         // GD: ↑
        {NEGLIGIBLE, HIGH, false, nullptr},            // DR: ——
        {MODERATE_DECREASE, HIGH, false, nullptr},     // CCB: ↓↓ — TGFβ brake disconnected in tumor
        {MODERATE_INCREASE, HIGH, true, "TGFβ now promotes invasion via SMAD-independent
MAPK/RhoA"},
        {CONTEXT_DEPENDENT, HIGH, true, "Tumor still secretes TGFβ (KRAS-driven), stroma still responds.
SMAD4 loss removes tumor's sensitivity, not the signal."},
        {MILD_INCREASE, MED, false, nullptr}           // IET: ↑
    },
    "THE MOST IMPORTANT GENE IN THE SIMULATION. Creates asymmetric TGFβ response: stroma
activates, tumor escapes. This asymmetry IS the stroma barrier problem."
},

// Gene 9: RB1 (functional inactivation via CDKN2A loss)
{
```

```
    GeneIndex::RB1, "FuncInact", CellTypeScope::TUMOR,
    {
        {MILD_INCREASE, HIGH, false, nullptr},        // GD: ↑
        {NEGLIGIBLE, HIGH, false, nullptr},           // DR: ——
        {MODERATE_DECREASE, HIGH, false, nullptr},    // CCB: ↓↓ — convergence node for all brake signals
        {NEGLIGIBLE, HIGH, false, nullptr},           // MI: ——
        {NEGLIGIBLE, HIGH, false, nullptr},           // PSS: ——
        {NEGLIGIBLE, HIGH, false, nullptr}            // IET: ——
    },
    "Downstream integrator. Model as output of CDKN2A/SMAD4/damage signals, not independent input."
},

// Gene 10: ZEB1 (transcriptional upregulation)
{
    GeneIndex::ZEB1, "Upregulated", CellTypeScope::TUMOR,
    {
        {MILD_DECREASE, LOW, true, "Go-vs-grow tradeoff: EMT cells migrate more, proliferate less"},
        {MILD_INCREASE, MED, false, nullptr},         // DR: ↑ — anoikis resistance
        {NEGLIGIBLE, HIGH, false, nullptr},           // CCB: ——
        {STRONG_INCREASE, HIGH, false, nullptr},      // MI: ↑↑↑ — EMT master switch
        {NEGLIGIBLE, HIGH, false, nullptr},           // PSS: ——
        {MODERATE_INCREASE, MED, false, nullptr}      // IET: ↑↑ — PD-L1 via CD274 promoter
    },
    "Activated by TGFβ (SMAD-independent) and HIF1A. Main output: CDH1=OFF. Enforce go-vs-grow penalty
on proliferation."
},

// Gene 11: CDH1 / E-cadherin (transcriptionally repressed by ZEB1)
{
    GeneIndex::CDH1, "Repressed", CellTypeScope::TUMOR,
    {
        {NEGLIGIBLE, HIGH, false, nullptr},           // GD: ——
        {NEGLIGIBLE, HIGH, false, nullptr},           // DR: ——
```

```
        {NEGLIGIBLE, HIGH, false, nullptr},          // CCB: ——
        {STRONG_INCREASE, HIGH, true, "Effect is when CDH1 is LOST: adhesion gone, motility unrestricted"},
        {NEGLIGIBLE, HIGH, false, nullptr},          // PSS: ——
        {NEGLIGIBLE, HIGH, false, nullptr}           // IET: ——
    },
    "Output node, not a driver. CDH1 state = readout of EMT status. Maps to cell_cell_adhesion_strength."
},

// Gene 12: MMP2 (transcriptional upregulation)
{
    GeneIndex::MMP2, "Upregulated", CellTypeScope::BOTH,
    {
        {NEGLIGIBLE, HIGH, false, nullptr},          // GD: ——
        {NEGLIGIBLE, HIGH, false, nullptr},          // DR: ——
        {NEGLIGIBLE, HIGH, false, nullptr},          // CCB: ——
        {STRONG_INCREASE, HIGH, false, nullptr},     // MI: ↑↑↑ — ECM degradation
        {MILD_DECREASE, MED, true, "Net stroma-degrading for Cycle 1. NOTE: MMP2 releases ECM-
sequestered TGFβ which can paradoxically increase stroma — flag for Cycle 2."},
        {NEGLIGIBLE, HIGH, false, nullptr}           // IET: ——
    },
    "Stroma-DEGRADING effector. EA may want to upregulate this. Cycle 2: add TGFβ liberation side effect."
},

// Gene 13: ACTA2 / α-SMA (upregulation in activated stellate cells)
{
    GeneIndex::ACTA2, "Upregulated", CellTypeScope::STROMA,
    {
        {MILD_INCREASE, MED, false, nullptr},        // GD: ↑ (stroma proliferation)
        {MILD_INCREASE, MED, false, nullptr},        // DR: ↑ (CAF apoptosis resistance)
        {NEGLIGIBLE, HIGH, false, nullptr},          // CCB: ——
        {MODERATE_INCREASE, HIGH, false, nullptr},   // MI: ↑↑ (contractile, motile)
        {STRONG_INCREASE, HIGH, false, nullptr},     // PSS: ↑↑↑ — operational definition of CAF activation
        {MILD_INCREASE, MED, false, nullptr}         // IET: ↑ — T cell exclusion
```

```
    },
    "STROMAL CELL ONLY. ACTA2=ON means PSC→myofibroblastic CAF. Outputs: COL1A1↑, HAS2↑, contractility↑."
},

// Gene 14: TGFB1 (overexpression by tumor and stroma)
{
    GeneIndex::TGFB1, "Overexpr", CellTypeScope::BOTH,
    {
        {CONTEXT_DEPENDENT, HIGH, true, "SMAD4-intact: GD↓ (growth arrest). SMAD4-null: GD── (brake disconnected)."},
        {NEGLIGIBLE, HIGH, false, nullptr},          // DR: ──
        {CONTEXT_DEPENDENT, HIGH, true, "SMAD4-intact: CCB↑↑ (arrest). SMAD4-null: CCB── (brake gone)."},
        {MODERATE_INCREASE, HIGH, true, "Via SMAD-independent arm (MAPK/RhoA). Active regardless of SMAD4."},
        {STRONG_INCREASE, HIGH, false, nullptr},     // PSS: ↑↑↑ — the barrier builder
        {STRONG_INCREASE, HIGH, false, nullptr}      // IET: ↑↑↑ — Treg induction, CD8 suppression
    },
    "DUAL-NATURE CYTOKINE. Effect on tumor depends on SMAD4 status. Effect on stroma is always activating. Model as diffusible substrate, not intracellular node."
},

// Gene 15: SHH (overexpression by tumor)
{
    GeneIndex::SHH, "Overexpr", CellTypeScope::TUMOR,
    {
        {NEGLIGIBLE, HIGH, false, nullptr},          // GD: ── (tumor doesn't respond to own SHH)
        {NEGLIGIBLE, HIGH, false, nullptr},          // DR: ──
        {NEGLIGIBLE, HIGH, false, nullptr},          // CCB: ──
        {NEGLIGIBLE, HIGH, false, nullptr},          // MI: ──
        {STRONG_INCREASE, HIGH, false, nullptr},     // PSS: ↑↑↑ — paracrine: tumor secretes, stroma responds
        {MILD_INCREASE, MED, false, nullptr}         // IET: ↑ — physical barrier to immune cells
```

```
    },
    "Purely paracrine. Vismodegib FAILED in PDAC — tumors accelerated. Stroma may also restrain tumor. Model must reproduce this."
},

// Gene 16: GLI1 (transcriptional activation in stroma)
{
    GeneIndex::GLI1, "Activated", CellTypeScope::STROMA,
    {
        {MODERATE_INCREASE, HIGH, false, nullptr},     // GD: ↑↑ (stroma proliferation)
        {MILD_INCREASE, MED, false, nullptr},          // DR: ↑ (stroma survival)
        {NEGLIGIBLE, HIGH, false, nullptr},            // CCB: ——
        {NEGLIGIBLE, HIGH, false, nullptr},            // MI: ——
        {MODERATE_INCREASE, HIGH, false, nullptr},     // PSS: ↑↑ — ECM gene expression
        {NEGLIGIBLE, HIGH, false, nullptr}             // IET: ——
    },
    "STROMAL CELL ONLY. Hedgehog effector TF. Separate from SHH so EA can target pathway at two levels."
},

// Gene 17: HAS2 (upregulation in activated CAFs)
{
    GeneIndex::HAS2, "Upregulated", CellTypeScope::STROMA,
    {
        {NEGLIGIBLE, HIGH, false, nullptr},            // GD: ——
        {NEGLIGIBLE, HIGH, false, nullptr},            // DR: ——
        {NEGLIGIBLE, HIGH, false, nullptr},            // CCB: ——
        {NEGLIGIBLE, HIGH, false, nullptr},            // MI: ——
        {STRONG_INCREASE, HIGH, false, nullptr},       // PSS: ↑↑↑ — HA increases interstitial pressure, blocks drug diffusion
        {MILD_INCREASE, MED, false, nullptr}           // IET: ↑
    },
    "STROMAL CELL ONLY. Produces hyaluronic acid. PEGPH20 failed Phase 3 (HALO-301) — HA removal alone insufficient."
```

```
},

// Gene 18: COL1A1 (upregulation in activated CAFs)
{
    GeneIndex::COL1A1, "Upregulated", CellTypeScope::STROMA,
    {
        {NEGLIGIBLE, HIGH, false, nullptr},          // GD: ——
        {NEGLIGIBLE, HIGH, false, nullptr},          // DR: ——
        {NEGLIGIBLE, HIGH, false, nullptr},          // CCB: ——
        {NEGLIGIBLE, HIGH, false, nullptr},          // MI: ——
        {STRONG_INCREASE, HIGH, false, nullptr},     // PSS: ↑↑↑ — fibrillar collagen, structural rigidity
        {MILD_INCREASE, MED, false, nullptr}         // IET: ↑ — dense collagen blocks T cell migration
    },
    "STROMAL CELL ONLY. Additive with HAS2 for ECM density field. Collagen highways can paradoxically
guide invasion — flag for Cycle 2."
},

// Gene 19: HIF1A (stabilization under hypoxia)
{
    GeneIndex::HIF1A, "Stabilized", CellTypeScope::BOTH,
    {
        {CONTEXT_DEPENDENT, HIGH, true, "Acute hypoxia slows growth; adapted cells upregulate glycolysis
and recover."},
        {MODERATE_INCREASE, HIGH, false, nullptr},   // DR: ↑↑ — BCL-XL upregulation, autophagy
        {NEGLIGIBLE, HIGH, false, nullptr},          // CCB: ——
        {MODERATE_INCREASE, HIGH, false, nullptr},   // MI: ↑↑ — direct ZEB1 activation
        {MODERATE_INCREASE, HIGH, false, nullptr},   // PSS: ↑↑ — CTGF, fibrosis signals
        {MODERATE_INCREASE, HIGH, false, nullptr}    // IET: ↑↑ — adenosine pathway, PD-L1
    },
    "Input node gated by local O2. Creates feedback loop: stroma→hypoxia→HIF1A→more stroma+resistance."
},

// Gene 20: NRF2 / NFE2L2 (gain-of-function or transcriptional activation)
```

```
{
    GeneIndex::NRF2, "GoF/Overexpr", CellTypeScope::TUMOR,
    {
        {MILD_INCREASE, MED, false, nullptr},          // GD: ↑ — metabolic reprogramming
        {MODERATE_INCREASE, MED, false, nullptr},      // DR: ↑↑ — drug metabolism + efflux
        {NEGLIGIBLE, HIGH, false, nullptr},            // CCB: ——
        {NEGLIGIBLE, HIGH, false, nullptr},            // MI: ——
        {NEGLIGIBLE, HIGH, false, nullptr},            // PSS: ——
        {NEGLIGIBLE, HIGH, false, nullptr}             // IET: ——
    },
    "Xenobiotic stress master regulator. Primary sim output: activates ABCB1. Growth effect is secondary, can be omitted to reduce complexity."
},

// Gene 21: ABCB1 / MDR1 (transcriptional upregulation)
{
    GeneIndex::ABCB1, "Upregulated", CellTypeScope::TUMOR,
    {
        {NEGLIGIBLE, HIGH, false, nullptr},            // GD: ——
        {STRONG_INCREASE, HIGH, true, "DRUG-CONDITIONAL: death resistance ONLY when drug substrate is present. Zero survival benefit at baseline."},
        {NEGLIGIBLE, HIGH, false, nullptr},            // CCB: ——
        {NEGLIGIBLE, HIGH, false, nullptr},            // MI: ——
        {NEGLIGIBLE, HIGH, false, nullptr},            // PSS: ——
        {NEGLIGIBLE, HIGH, false, nullptr}             // IET: ——
    },
    "Drug efflux pump. Effect = ABCB1 * drug_present → intracellular_drug↓↓. Map to drug uptake modifier, NOT basal apoptosis rate."
}

UTILITY FUNCTIONS:

// Get the effect profile for a gene
```

```
const GeneBehaviorProfile& get_gene_profile(GeneIndex gene);

// Get the raw directional effect of a gene on an axis
EffectDirection get_effect(GeneIndex gene, BehaviorAxis axis);

// Convert EffectDirection to a numeric multiplier for use in phenotype calculations
// Maps: ↑↑↑→+1.0, ↑↑→+0.66, ↑→+0.33, ——→0.0, ↓→-0.33, ↓↓→-0.66, ↓↓↓→-1.0
// CONTEXT_DEPENDENT returns 0.0 (must be handled by conditional logic elsewhere)
double effect_to_multiplier(EffectDirection dir);

// Check if a gene's effect on an axis requires conditional logic
bool requires_conditional(GeneIndex gene, BehaviorAxis axis);

// Print the full matrix (for debugging)
void print_effect_matrix();

Use constexpr where possible. Include static_assert to verify array sizes.
Add a block comment at the top explaining the effect scale and confidence levels.
```

Phase B: Refined Boolean Network with Multi-Axis Effects

Prompt B.1 — SMAD4 Asymmetry (Critical)

```
IMPORTANT CONTEXT: This prompt modifies the Boolean network from the base Stroma World roadmap
(Prompt 1.2).
The SMAD4 asymmetry is THE most important biological feature of the entire simulation.

Modify boolean_network.cpp to implement the SMAD4-dependent TGF-β response asymmetry.

THE PROBLEM:
TGF-β is a diffusible cytokine that has OPPOSITE effects depending on which cell receives it:
```

- Tumor cell WITH intact SMAD4: TGF-β → growth arrest + apoptosis (tumor suppressor)
- Tumor cell WITHOUT SMAD4 (most PDAC): TGF-β → invasion via SMAD-independent pathways (no arrest)
- Stromal cell (always SMAD4-intact): TGF-β → full CAF activation → ECM production

In standard PDAC (SMAD4 lost), the tumor secretes TGF-β that activates stroma but cannot brake itself.
The tumor builds its own prison. This asymmetry IS the stroma barrier problem.

IMPLEMENTATION:

In the tumor cell Boolean rules, replace any TGF-β-related rules with this conditional block:

```
void BooleanNetwork::compute_tgfb_response_tumor(double tgfb_local) {
    double smad4 = gene_states[GeneIndex::SMAD4];

    // === SMAD4-DEPENDENT (canonical) pathway ===
    // Only active when SMAD4 is present
    double smad4_signal = smad4 * tgfb_local;

    // Growth arrest: TGF-β brakes proliferation via SMAD4→p21→RB
    // When SMAD4 is lost (=0), this entire branch is dead.
    double tgfb_growth_arrest = smad4_signal * 0.6;

    // Apoptosis induction: TGF-β can trigger apoptosis via SMAD4→BIM
    double tgfb_apoptosis = smad4_signal * 0.3;

    // === SMAD-INDEPENDENT (non-canonical) pathways ===
    // These are ALWAYS active regardless of SMAD4 status.
    // In SMAD4-null cells, these are the ONLY TGF-β effects.
    double tgfb_noncanonical = tgfb_local;  // full signal

    // Invasion promotion: TGF-β → MAPK → RhoA → motility
    double tgfb_invasion = tgfb_noncanonical * 0.4;
```

```
    // EMT promotion: TGF-β → ZEB1 (SMAD-independent arm)
    double tgfb_emt = tgfb_noncanonical * 0.3;

    // === Apply to target states ===

    // RB1 target: add SMAD4-dependent brake contribution
    // (This is in addition to CDKN2A contribution from base rules)
    target[GeneIndex::RB1] += tgfb_growth_arrest * 0.3;

    // BAX target: add SMAD4-dependent pro-apoptosis contribution
    target[GeneIndex::BAX] += tgfb_apoptosis;

    // ZEB1 target: add SMAD-independent invasion/EMT contribution
    // NOTE: This works in SMAD4-null cells (the common PDAC case)
    target[GeneIndex::ZEB1] += tgfb_emt;

    // MMP2 target: TGF-β promotes ECM remodeling via non-canonical
    target[GeneIndex::MMP2] += tgfb_invasion * 0.3;

    // Clamp all targets to [0, 1]
    for (int i = 0; i < GENE_COUNT; i++) {
        target[i] = std::clamp(target[i], 0.0, 1.0);
    }
}
```

For stromal cells, TGF-β response is unconditional (SMAD4 always intact):

```
void BooleanNetwork::compute_tgfb_response_stroma(double tgfb_local) {
    // Stromal cells always have intact SMAD4.
    // TGF-β is the primary CAF activation signal.

    // ACTA2 activation (PSC → myofibroblast)
    target[GeneIndex::ACTA2] += tgfb_local * 0.6;
```

```
    // ECM production (downstream of ACTA2, but TGF-β also has direct effect)
    target[GeneIndex::COL1A1] += tgfb_local * 0.2;
    target[GeneIndex::HAS2] += tgfb_local * 0.15;

    // Paracrine feedback: activated CAFs secrete MORE TGF-β
    target[GeneIndex::TGFB1] += gene_states[GeneIndex::ACTA2] * 0.4;

    // Clamp
    for (int i = 0; i < GENE_COUNT; i++) {
        target[i] = std::clamp(target[i], 0.0, 1.0);
    }
}
```

VERIFICATION TEST:
Write a standalone test that creates two tumor cells — one with SMAD4=1.0 (intact), one with SMAD4=0.0 (lost) — exposes both to tgfb_local=0.8, runs 100 Boolean update steps, and verifies:
- SMAD4-intact cell: RB1 > 0.5 (growth arrested), BAX elevated
- SMAD4-null cell: RB1 stays low, ZEB1 elevated, MMP2 elevated
- Both cells: TGFB1 secretion is similar (KRAS-driven, not SMAD4-dependent)

This test is the SINGLE MOST IMPORTANT validation in the project. If it fails, the stroma barrier will not emerge.

Add detailed block comments explaining the biological rationale at every step.
Cite the relevant PDAC literature concepts: canonical vs non-canonical TGF-β signaling.

Prompt B.2 — Go-vs-Grow Tradeoff Enforcement

Modify the tumor cell Boolean network and phenotype mapping to enforce the go-vs-grow tradeoff.

BIOLOGICAL RATIONALE:
EMT (ZEB1=ON, mesenchymal phenotype) and maximal proliferation (MYC high) are partially antagonistic.

Cells in active migration invest energy in cytoskeletal remodeling and motility machinery rather than DNA replication. A cell should not be simultaneously maximally migratory AND maximally proliferative.

IMPLEMENTATION (in boolean_network.cpp, tumor rules):

```
// After computing all target states, apply the go-vs-grow penalty:
void BooleanNetwork::apply_go_vs_grow(double dt) {
    double zeb1 = gene_states[GeneIndex::ZEB1];
    double myc = gene_states[GeneIndex::MYC];

    // Penalty increases with the product of ZEB1 and MYC
    // Max penalty when both are at 1.0
    double conflict = zeb1 * myc;

    if (conflict > 0.3) {
        // In conflict: reduce BOTH slightly (biological: cells must "choose")
        // MYC gets a motility-dependent penalty
        double myc_penalty = 0.3 * zeb1;  // up to 30% MYC reduction when fully mesenchymal
        target[GeneIndex::MYC] *= (1.0 - myc_penalty);

        // ZEB1 gets a mild proliferation-dependent penalty (cells that are dividing fast are less motile)
        double zeb1_penalty = 0.15 * myc;
        target[GeneIndex::ZEB1] *= (1.0 - zeb1_penalty);
    }
}
```

ALSO modify tumor_cell.cpp phenotype mapping (Prompt 2.1 from base roadmap):

```
// In the proliferation rate calculation, add ZEB1-dependent penalty:
double go_vs_grow_factor = 1.0 - 0.3 * bn->gene(GeneIndex::ZEB1);
// A fully mesenchymal cell (ZEB1=1.0) proliferates at 70% of its epithelial rate

phenotype.cycle.data.transition_rate(0, 0) =
```

```
    base_prolif_rate * growth_drive * brake * go_vs_grow_factor;

// In the motility calculation, add a subtle inverse effect:
// Highly proliferative cells are slightly less motile
double grow_vs_go_factor = 1.0 - 0.15 * bn->gene(GeneIndex::MYC);
if (is_mesenchymal) {
    phenotype.motility.migration_speed = 1.0 * grow_vs_go_factor;
}
```

VERIFICATION:
Create test scenarios:
1. High MYC + Low ZEB1 → fast proliferation, slow motility (epithelial grower)
2. Low MYC + High ZEB1 → slow proliferation, fast motility (mesenchymal invader)
3. High MYC + High ZEB1 → both penalized, intermediate state (biological: this state is unstable)
Log the actual proliferation rates and motility speeds and confirm the tradeoff.

Prompt B.3 — Drug-Conditional ABCB1 Logic

Modify the Boolean network and phenotype mapping to implement drug-conditional ABCB1 behavior.

BIOLOGICAL RATIONALE:
ABCB1 (MDR1) is a drug efflux pump. It physically pumps chemotherapy out of the cell.
It provides ZERO survival advantage in the absence of drug. It is an INDUCED resistance
mechanism: drug exposure → NRF2 activation → ABCB1 upregulation → drug efflux.

IMPLEMENTATION:

1. In boolean_network.cpp, the ABCB1 target rule already depends on NRF2 and HIF1A.
   The NRF2 rule already depends on drug presence. This is correct.
   VERIFY the chain: drug_local > threshold → NRF2↑ → ABCB1↑

2. In tumor_cell.cpp phenotype mapping, replace the drug sensitivity section with:

```
// === DRUG-CONDITIONAL ABCB1 EFFECT ===
double nrf2 = bn->gene(GeneIndex::NRF2);
double abcb1 = bn->gene(GeneIndex::ABCB1);
double drug_local = pCell->nearest_density_vector()[drug_index];

// Base drug sensitivity (before efflux)
double base_sensitivity = 1.0 - 0.4 * nrf2;  // NRF2 provides metabolic resistance

// ABCB1 effect is MULTIPLICATIVE with drug presence
// When drug=0, ABCB1 does nothing. When drug>0, ABCB1 reduces effective drug.
double efflux_factor;
if (drug_local > 0.01) {
    // ABCB1 actively pumping: reduces intracellular drug concentration
    efflux_factor = 1.0 - 0.6 * abcb1;  // up to 60% drug removal when ABCB1=1.0
} else {
    // No drug present: ABCB1 has no effect on survival
    efflux_factor = 1.0;
}

double effective_drug_sensitivity = base_sensitivity * efflux_factor;
effective_drug_sensitivity = std::max(0.05, effective_drug_sensitivity);  // floor: never fully immune
pCell->custom_data["drug_sensitivity"] = effective_drug_sensitivity;

// Drug-induced death: only when drug is present AND reaches the cell
if (drug_local > 0.01) {
    double intracellular_drug = drug_local * effective_drug_sensitivity;
    // Drug effect on apoptosis is proportional to intracellular concentration
    double drug_death_boost = intracellular_drug * 0.002;  // tunable parameter
    phenotype.death.rates[apoptosis_index] += drug_death_boost;
}

// === DO NOT add ABCB1 to basal apoptosis calculation ===
// The basal apoptosis rate (BCL-XL/BAX ratio, TP53 status) is INDEPENDENT of ABCB1.
```

```
// ABCB1 only affects drug-induced death. This distinction is critical.

3. Add a custom data field "intracellular_drug" to track effective drug concentration per cell.
   This is needed by the output parser for the fitness function.

VERIFICATION:
Run two scenarios:
  A. Drug present (drug boundary = 1.0), ABCB1 starts low → observe: tumor death initially,
     then survivors upregulate NRF2/ABCB1, death rate decreases over time (induced resistance)
  B. No drug (drug boundary = 0.0), ABCB1 artificially set high → observe: ZERO difference
     in survival compared to ABCB1=0. Confirm ABCB1 is truly drug-conditional.
```

Prompt B.4 — Multi-Axis KRAS Effects

```
Modify the tumor cell Boolean network to capture KRAS as the root node that drives ALL six axes,
not just growth. The current base roadmap Boolean rules underrepresent KRAS's cross-axis effects.

KRAS (constitutively active) should contribute to:
  - Growth Drive: ↑↑↑ (already modeled: KRAS→MYC→proliferation)
  - Death Resistance: ↑↑ (ADD: KRAS→PI3K/AKT→BCL-XL upregulation)
  - Cell Cycle Braking: ↓↓ (ADD: KRAS signaling overrides some checkpoint signals)
  - Motility/Invasion: ↑ (ADD: KRAS→RalGEF→RAC→motility)
  - Pro-Stroma Signaling: ↑↑↑ (already modeled: KRAS drives TGFB1 and SHH secretion)
  - Immune Evasion: ↑↑ (tracked but not simulated in Cycle 1)

ADDITIONS to the tumor Boolean rules in boolean_network.cpp:

// Death resistance contribution from KRAS (via PI3K/AKT pathway)
// This is in addition to the existing BCL-XL rule
target[GeneIndex::BCL_XL] = 0.3 + 0.5 * gene_states[GeneIndex::KRAS]
                            + 0.2 * gene_states[GeneIndex::NRF2];
// ↑ Already exists in base roadmap. Confirm the 0.5*KRAS coefficient is present.
```

```
// Cell cycle braking override from KRAS
// KRAS signaling partially overrides RB1-mediated arrest even when RB1 is active
// This models KRAS's direct effect on CDK4/6 activity via cyclin D1
double kras_brake_override = 0.3 * gene_states[GeneIndex::KRAS];
// Applied in the phenotype mapping:
// brake factor = max(brake, kras_brake_override)  — KRAS sets a floor on proliferation

// Motility contribution from KRAS (RalGEF→RAC pathway)
// In the ZEB1 rule, KRAS already contributes 0.2. Verify this is present.
// ALSO add a direct motility boost in the phenotype mapping:
// Even epithelial cells with active KRAS have slightly elevated motility
double kras_motility_boost = 0.15 * gene_states[GeneIndex::KRAS];
// Applied: motility_speed += kras_motility_boost (capped at max)

// Pro-stroma secretion from KRAS (already modeled via TGFB1 and SHH rules)
// Verify: TGFB1 target = 0.3*KRAS + ... and SHH target = 0.3*KRAS + ...
// These coefficients encode KRAS ↑↑↑ on PSS.

Add all contributions as clearly commented blocks with the axis name and biological pathway noted.
Make sure each KRAS contribution can be individually toggled with a #define or constexpr bool
for ablation studies (e.g., "what happens if we remove KRAS→motility but keep KRAS→growth?").
```

Prompt B.5 — MYC Conditional Death Resistance

```
Implement the conditional MYC → death resistance logic.

BIOLOGICAL RATIONALE:
MYC has a PARADOXICAL effect on apoptosis:
  - In TP53-INTACT cells: MYC SENSITIZES to apoptosis (oncogene-induced apoptosis via ARF→p53)
  - In TP53-NULL cells (most PDAC): MYC is mildly pro-survival (the safety brake is gone)

This is one of the key reasons PDAC is so hard to kill: the combination of MYC amplification +
TP53 loss removes the safety valve that would normally cause MYC-overexpressing cells to die.
```

IMPLEMENTATION in boolean_network.cpp, tumor rules:

```
// MYC's effect on death axis depends on TP53 status
double tp53 = gene_states[GeneIndex::TP53];
double myc = gene_states[GeneIndex::MYC];

double myc_death_effect;
if (tp53 > 0.5) {
    // TP53 intact: MYC sensitizes to apoptosis (oncogene-induced apoptosis)
    // This is the normal safety mechanism
    myc_death_effect = -0.2 * myc;  // negative = pro-death
} else {
    // TP53 lost: MYC is mildly pro-survival
    // ARF→p53→apoptosis pathway is broken, MYC drives metabolic fitness
    myc_death_effect = 0.1 * myc;   // positive = pro-survival
}

// Apply to BCL-XL target (modulates death resistance)
target[GeneIndex::BCL_XL] += myc_death_effect;

// Apply to BAX target (inverse direction)
target[GeneIndex::BAX] -= myc_death_effect * 0.5;

// Clamp
target[GeneIndex::BCL_XL] = std::clamp(target[GeneIndex::BCL_XL], 0.0, 1.0);
target[GeneIndex::BAX] = std::clamp(target[GeneIndex::BAX], 0.0, 1.0);
```

VERIFICATION:
Test with 3 cell configurations:
  A. MYC=1.0, TP53=1.0 → BAX should be elevated (MYC-induced apoptosis active)
  B. MYC=1.0, TP53=0.0 → BCL-XL slightly elevated, BAX low (PDAC state: survival)
  C. MYC=0.0, TP53=0.0 → baseline (neither MYC-driven death nor MYC-driven survival)

Prompt B.6 — Cell Type Scope Enforcement

Add a compile-time and runtime safety system to prevent stroma-only genes from being expressed in tumor cells and vice versa.

PROBLEM: If ACTA2, GLI1, HAS2, or COL1A1 accidentally get activated in tumor cells, the simulation will produce tumor cells that build their own stroma — biologically wrong. Similarly, genes like KRAS (mutant), TP53, CDKN2A, SMAD4 should not be mutant/locked in stromal cells.

IMPLEMENTATION:

1. In boolean_network.cpp, add scope enforcement to the update() method:

```
void BooleanNetwork::enforce_cell_type_scope() {
    for (int i = 0; i < GENE_COUNT; i++) {
        CellTypeScope scope = GENE_INFO[i].cell_type;

        if (cell_type_ == CellType::TUMOR && scope == CellTypeScope::STROMA) {
            // This gene should NOT be active in tumor cells
            gene_states[i] = 0.0;
            target[i] = 0.0;
        }
        if (cell_type_ == CellType::STROMA && scope == CellTypeScope::TUMOR) {
            // This gene should NOT be active in stromal cells
            // Exception: some tumor-only genes have a baseline expression in stroma
            // but their MUTANT forms (locked states) should not carry over
            if (is_mutant[i]) {
                is_mutant[i] = false;  // stroma doesn't carry tumor mutations
                gene_states[i] = GENE_INFO[i].default_stroma_state;
            }
        }
    }
```

```
}

// Call this at the END of every update():
void BooleanNetwork::update(double dt, double oxygen, double tgfb_local, double shh_local, double drug_local) {
    compute_targets(oxygen, tgfb_local, shh_local, drug_local);
    apply_continuous_update(dt);
    enforce_cell_type_scope();  // <-- always last
}
```

2. Add a compile-time check in gene_definitions.h:

```
// Genes that must NEVER be active in tumor cells:
static_assert(GENE_INFO[GeneIndex::ACTA2].cell_type == CellTypeScope::STROMA);
static_assert(GENE_INFO[GeneIndex::GLI1].cell_type == CellTypeScope::STROMA);
static_assert(GENE_INFO[GeneIndex::HAS2].cell_type == CellTypeScope::STROMA);
static_assert(GENE_INFO[GeneIndex::COL1A1].cell_type == CellTypeScope::STROMA);
```

3. Add a runtime assertion in the phenotype update functions:

```
// At the top of tumor_phenotype_update():
assert(pCell->type == tumor_cell_type_index && "tumor_phenotype_update called on non-tumor cell!");

// At the top of stromal_phenotype_update():
assert(pCell->type == stromal_cell_type_index && "stromal_phenotype_update called on non-stromal cell!");
```

4. Add a debug logging mode that prints a WARNING if any gene fires outside its scope.

Phase C: Enhanced Phenotype Mapping with Confidence Weighting

Prompt C.1 — Confidence-Weighted Parameter Scaling

```
Create custom_modules/confidence_scaling.h that uses the confidence levels from the behavior
map to determine how strongly each gene→phenotype connection should be weighted.

RATIONALE:
HIGH confidence effects should map to strong phenotype changes.
MED confidence effects should be attenuated (we're less sure about the magnitude).
LOW confidence effects should be very weak (included for completeness, not driving behavior).

IMPLEMENTATION:

constexpr double CONFIDENCE_SCALE[] = {
    1.0,   // HIGH — full effect
    0.6,   // MED — attenuated to 60%
    0.3    // LOW — weak, 30%
};

// Usage in phenotype mapping:
// Instead of: motility_speed = 1.0 * zeb1;
// Use: motility_speed = 1.0 * zeb1 * get_confidence_scale(GeneIndex::ZEB1,
BehaviorAxis::MOTILITY_INVASION);

double get_confidence_scale(GeneIndex gene, BehaviorAxis axis) {
    const GeneBehaviorProfile& profile = GENE_PROFILES[gene];
    Confidence conf = profile.effects[axis].confidence;
    return CONFIDENCE_SCALE[static_cast<int>(conf)];
}

Apply this scaling to ALL gene→phenotype mappings in tumor_cell.cpp and stromal_cell.cpp.
```

This creates a natural attenuation: well-established effects (KRAS→growth, BCL-XL→survival) dominate behavior, while uncertain effects (EGFR→motility, NRF2→growth) are present but subdued.

Also add a config option to DISABLE confidence scaling (set all to 1.0) for sensitivity analysis.

Phase D: BCL-XL Toxicity Cost in Fitness Function

Prompt D.1 — Intervention Toxicity Penalty

Modify python/ea/fitness.py to add a toxicity penalty for BCL-XL inhibition.

BIOLOGICAL RATIONALE:
Navitoclax (BCL-XL inhibitor) causes severe thrombocytopenia (platelet destruction) because platelets depend on BCL-XL for survival. This is a dose-limiting toxicity that prevents full BCL-XL inhibition in patients. The EA must account for this or it will discover "solutions" that are clinically infeasible.

IMPLEMENTATION:

Add a new fitness component:

```
def compute_toxicity_penalty(individual: list[dict]) -> float:
    """
    Penalizes interventions with known clinical toxicity profiles.
    Returns a penalty value in [0, 1] where 0 = no toxicity, 1 = lethal toxicity.
    """
    penalty = 0.0

    for intervention in individual:
        gene = intervention["gene"]
        strength = intervention["strength"]
```

```
        # BCL-XL inhibition → thrombocytopenia
        # Toxicity scales sharply above 50% inhibition
        if gene == "BCL_XL" and intervention["effect"] == "INHIBIT":
            if strength > 0.5:
                # Exponential penalty above 50% inhibition
                penalty += 0.3 * ((strength - 0.5) / 0.5) ** 2
            else:
                penalty += 0.05 * strength  # mild at low doses

        # SHH inhibition → potential tumor acceleration (vismodegib failure)
        # This is captured by the invasion penalty, but add a warning flag
        if gene == "SHH" and intervention["effect"] == "INHIBIT" and strength > 0.7:
            penalty += 0.1  # mild penalty as a prior against aggressive SHH targeting

    return min(1.0, penalty)
```

Update the total fitness calculation:

```
fitness = (0.30 * tumor_kill
         + 0.25 * stroma_disruption
         + 0.20 * penetration
         - 0.10 * resistance_penalty
         - 0.10 * invasion_penalty
         - 0.05 * toxicity_penalty)   # <-- NEW

# Weights reduced slightly from base roadmap to accommodate toxicity term.
# Total positive weights: 0.75. Total penalty weights: 0.25.
```

Also add a function to flag interventions that map to known clinical failures:

```
def check_clinical_feasibility(individual: list[dict]) -> list[str]:
    """Returns a list of warning strings for clinically problematic strategies."""
```

```
    warnings = []
    genes_targeted = {i["gene"] for i in individual}

    if "SHH" in genes_targeted and len(genes_targeted) == 1:
        warnings.append("WARNING: SHH monotherapy failed in PDAC (vismodegib trial). "
                        "Tumors may accelerate without stroma restraint.")

    if "HAS2" in genes_targeted and len(genes_targeted) == 1:
        warnings.append("WARNING: PEGPH20 monotherapy failed Phase 3 (HALO-301). "
                        "HA removal alone is insufficient.")

    if "BCL_XL" in genes_targeted:
        bcl_strength = next(i["strength"] for i in individual if i["gene"] == "BCL_XL")
        if bcl_strength > 0.8:
            warnings.append(f"WARNING: BCL-XL inhibition at {bcl_strength:.0%} likely causes "
                            "severe thrombocytopenia. May not be clinically achievable.")

    return warnings
```

Phase E: Clinical Trial Failure Validation Suite

Prompt E.1 — Reproduce Known Failures

```
Create python/validation/clinical_trial_validation.py

This is the MOST IMPORTANT validation file in the project. If the simulation cannot reproduce
known clinical trial failures, the gene→behavior mapping has bugs.

THREE MANDATORY VALIDATION SCENARIOS:

=== SCENARIO 1: VISMODEGIB FAILURE (SHH inhibition accelerates tumor) ===
```

```
def test_vismodegib_failure(runner: PhysiCellRunner, parser: OutputParser):
    """
    Vismodegib (SHH pathway inhibitor) failed in PDAC clinical trials.
    Tumors actually ACCELERATED when stroma was depleted via SHH inhibition.

    Hypothesis: Stroma physically restrains tumor growth. Remove stroma → tumor expands.

    Test:
    1. Run baseline (no intervention) → record tumor_count_baseline, tumor_extent_baseline
    2. Run with SHH INHIBIT at strength 0.9 → record tumor_count_shh, tumor_extent_shh
    3. ASSERT: tumor_extent_shh > tumor_extent_baseline (tumor spread more with stroma gone)
    4. ASSERT: stroma_barrier_score_shh < stroma_barrier_score_baseline (stroma was indeed reduced)
    5. OPTIONAL: tumor_count_shh >= tumor_count_baseline (not killed, possibly grew faster)

    If assertion 3 fails: The model doesn't capture stroma-as-restraint. Check if:
      - ECM density affects tumor cell motility (dense ECM should slow invasion)
      - Stromal cells physically block tumor cell migration (volume exclusion)
      - CAF-derived signals have any direct anti-tumor effects

    PASS CRITERIA:
      tumor_extent_shh / tumor_extent_baseline > 1.1 (at least 10% more invasion)
      AND stroma_barrier_score_shh < 0.5 * stroma_barrier_score_baseline
    """
    # Run baseline
    baseline_result = runner.run(generate_no_intervention_json())
    baseline_metrics = parser.parse_final_state(baseline_result.run_dir)

    # Run vismodegib analog
    vismodegib = [{"gene": "SHH", "effect": "INHIBIT", "strength": 0.9, "name": "vismodegib"}]
    vismo_result = runner.run(generate_intervention_json(vismodegib))
    vismo_metrics = parser.parse_final_state(vismo_result.run_dir)
```

```
    # Assertions
    invasion_ratio = vismo_metrics.tumor_extent / baseline_metrics.tumor_extent
    stroma_ratio = vismo_metrics.stroma_barrier_score / baseline_metrics.stroma_barrier_score

    result = {
        "scenario": "VISMODEGIB_FAILURE",
        "invasion_ratio": invasion_ratio,
        "stroma_ratio": stroma_ratio,
        "pass": invasion_ratio > 1.1 and stroma_ratio < 0.5,
        "interpretation": (
            "PASS: Model reproduces vismodegib failure. Stroma removal increased invasion."
            if invasion_ratio > 1.1 else
            "FAIL: Model does NOT reproduce vismodegib failure. Stroma removal did not increase invasion. "
            "Check ECM→motility coupling and volume exclusion mechanics."
        )
    }
    return result


=== SCENARIO 2: PEGPH20 PARTIAL FAILURE (HA removal insufficient alone) ===

def test_pegph20_insufficiency(runner, parser):
    """
    PEGPH20 (hyaluronidase targeting HAS2-produced HA) showed initial promise
    but failed Phase 3 (HALO-301). HA removal alone is not enough because:
    - Collagen barrier remains
    - Immune evasion remains
    - Tumor resistance mechanisms are untouched

    Test:
    1. Run baseline
    2. Run with HAS2 INHIBIT at 0.9 → partial barrier removal
    3. Run with HAS2 INHIBIT 0.9 + EGFR INHIBIT 0.7 → combination
```

```
    4. ASSERT: HAS2-only improves drug penetration but does NOT significantly reduce tumor
    5. ASSERT: Combination is better than either alone

    PASS CRITERIA:
      drug_penetration_has2 > drug_penetration_baseline * 1.3 (HA removal helps penetration)
      AND tumor_kill_has2 < 0.3 (but doesn't kill tumor alone — insufficient monotherapy)
      AND fitness_combination > fitness_has2_only (combination needed)
    """
    # ... implementation follows same pattern as scenario 1


=== SCENARIO 3: COMBINATION SYNERGY REQUIREMENT ===

def test_combination_synergy(runner, parser):
    """
    Nab-paclitaxel works in PDAC only in combination with gemcitabine. This models the general
    principle: stroma disruption requires COMBINATION with cytotoxic therapy.

    Test:
    1. Run stroma-targeting only: TGFB1 INHIBIT 0.7 + HAS2 INHIBIT 0.7
    2. Run drug only: high drug boundary, no stroma targeting
    3. Run combination: stroma targeting + drug
    4. ASSERT: Combination fitness > sum of individual fitnesses (synergy, not just additivity)

    PASS CRITERIA:
      fitness_combo > 1.2 * max(fitness_stroma_only, fitness_drug_only)
    """
    # ... implementation


=== RUNNER ===

def run_all_clinical_validations(runner, parser):
```

```
    """Run all three validation scenarios and produce a report."""
    results = []
    results.append(test_vismodegib_failure(runner, parser))
    results.append(test_pegph20_insufficiency(runner, parser))
    results.append(test_combination_synergy(runner, parser))

    print("\n=== CLINICAL TRIAL VALIDATION REPORT ===")
    all_pass = True
    for r in results:
        status = "✓ PASS" if r["pass"] else "✗ FAIL"
        print(f"  {status}  {r['scenario']}")
        print(f"         {r['interpretation']}")
        if not r["pass"]:
            all_pass = False

    if all_pass:
        print("\nAll clinical validations passed. Gene→behavior mapping is biologically plausible.")
    else:
        print("\nSOME VALIDATIONS FAILED. Gene→behavior mapping needs debugging.")
        print("The failed scenarios indicate which biological mechanisms are incorrectly modeled.")

    return results
```

Include detailed docstrings for every function. Save results to JSON for tracking across model versions.
This validation suite should be run after EVERY change to the Boolean network or phenotype mapping.

Phase F: Stroma-as-Restraint Mechanics

Prompt F.1 — ECM Physical Restraint of Tumor

The vismodegib validation (Phase E) requires that stroma physically restrains tumor expansion.

Without this mechanic, removing stroma cannot accelerate the tumor, and the validation will fail.

Add ECM-dependent motility restriction to tumor_cell.cpp:

In the motility section of tumor_phenotype_update():

```
// === ECM-DEPENDENT MOTILITY ===
// Dense ECM RESTRICTS tumor cell movement (physical barrier).
// This is distinct from drug diffusion blocking — it's about cell migration through matrix.
double local_ecm = pCell->nearest_density_vector()[ecm_index];

// Two competing effects of ECM on motility:
// 1. LOW ECM: cells move freely (no barrier)
// 2. MODERATE ECM: cells are somewhat impeded
// 3. HIGH ECM: cells are physically trapped (dense desmoplasia)
// 4. BUT: MMP2 degrades ECM, creating paths

double ecm_motility_factor;
if (local_ecm < 0.2) {
    // Low ECM: free movement
    ecm_motility_factor = 1.0;
} else if (local_ecm < 0.6) {
    // Moderate ECM: linear impedance
    ecm_motility_factor = 1.0 - 0.5 * (local_ecm - 0.2) / 0.4;
} else {
    // Dense ECM: severe restriction
    ecm_motility_factor = 0.5 - 0.4 * (local_ecm - 0.6) / 0.4;
    ecm_motility_factor = std::max(0.1, ecm_motility_factor);  // floor: never fully immobile
}

// MMP2 partially counteracts ECM restriction (cell carves its own path)
double mmp2_relief = 0.3 * bn->gene(GeneIndex::MMP2);
ecm_motility_factor = std::min(1.0, ecm_motility_factor + mmp2_relief);
```

```
// Apply to final motility
phenotype.motility.migration_speed *= ecm_motility_factor;

ALSO add ECM-dependent proliferation restriction:
// Dense stroma compresses tumor cells (solid stress), reducing proliferation
double solid_stress_factor = 1.0 - 0.3 * std::max(0.0, local_ecm - 0.5);
phenotype.cycle.data.transition_rate(0, 0) *= solid_stress_factor;

// This creates the mechanic where stroma restrains the tumor:
// Remove stroma → ecm_motility_factor → 1.0, solid_stress_factor → 1.0
// → Tumor proliferates faster AND migrates faster
// → Vismodegib scenario produces accelerated, spreading tumor
// → Validation PASSES

Add detailed comments explaining the physical biology:
- Interstitial fluid pressure from HA compresses vasculature
- Dense collagen creates physical barriers to cell migration
- Solid stress from expanding stroma compresses tumor nests
- MMP2 creates proteolytic paths through ECM
```

Phase G: Context-Dependent (◆) Logic Summary

Prompt G.1 — Implement All Context-Dependent Entries

```
The behavior map has four genes with context-dependent (◆) entries that require conditional
logic rather than static multipliers. Verify ALL four are correctly implemented:

1. KRAS × Growth Drive: NOT context-dependent in Cycle 1 (always ↑↑↑, constitutively active).
   ✓ No action needed.
```

2. SMAD4 × Pro-Stroma Signaling: ◆
IMPLEMENTED IN: Prompt B.1 (SMAD4 asymmetry)
VERIFY: Tumor cells with SMAD4=0 still secrete TGF-β at the same rate as SMAD4=1 cells.
The difference is response, not secretion. Run test: two tumors, one SMAD4+, one SMAD4-.
Both should produce similar TGF-β fields. Stroma activation should be similar.
But SMAD4- tumor should grow faster (no brake) and invade more (SMAD-independent EMT).

3. TGFB1 × Growth Drive and Cell Cycle Braking: ◆
IMPLEMENTED IN: Prompt B.1 (compute_tgfb_response_tumor)
VERIFY: Same as #2. TGF-β's effect on growth and braking depends on SMAD4.

4. HIF1A × Growth Drive: ◆
ISSUE: Acute hypoxia slows growth, but HIF1A-adapted cells recover via glycolysis.
IMPLEMENT TIME-DEPENDENT HIF1A RESPONSE:

```
double hif1a = gene_states[GeneIndex::HIF1A];
double hif1a_duration = pCell->custom_data["hif1a_active_minutes"];

double hif1a_growth_effect;
if (hif1a > 0.5) {
    hif1a_active_minutes += dt;
    if (hif1a_active_minutes < 360.0) {
        // Acute phase (first 6 hours): growth INHIBITED
        hif1a_growth_effect = -0.3 * hif1a;
    } else {
        // Adapted phase: growth partially recovers via metabolic switch
        double adaptation = std::min(1.0, (hif1a_active_minutes - 360.0) / 1440.0);
        hif1a_growth_effect = -0.3 * hif1a * (1.0 - 0.7 * adaptation);
        // At full adaptation: only 30% of growth penalty remains
    }
} else {
    hif1a_active_minutes = 0.0;  // reset counter when normoxic
    hif1a_growth_effect = 0.0;
```

```
}

// Apply to proliferation rate
phenotype.cycle.data.transition_rate(0, 0) *= (1.0 + hif1a_growth_effect);

Add "hif1a_active_minutes" to tumor cell custom data fields.

FINAL CHECK: Grep the entire codebase for any place where a ◆ gene×axis combination
is treated as a static multiplier. All four must use conditional logic. Print a warning
during initialization if any GeneBehaviorProfile with CONTEXT_DEPENDENT effects doesn't
have a corresponding conditional handler registered.
```

Appendix D: Complete Effect Matrix as Code-Ready Lookup

```
FOR REFERENCE WHEN IMPLEMENTING ANY GENE→PHENOTYPE CONNECTION.
Copy this table into code comments wherever gene effects are applied.

TUMOR CELLS:
Gene       | GD   | DR   | CCB  | MI   | PSS  | IET  | Key conditional
-----------|------|------|------|------|------|------|------------------
KRAS       | +1.0 | +0.66| -0.66| +0.33| +1.0 | +0.66| None (always ON)
MYC        | +1.0 | ◆    | -1.0 | 0    | +0.33| +0.33| DR depends on TP53
EGFR       | +0.66| +0.33| -0.33| +0.33| 0    | 0    | None
TP53(lost) | +0.33| +1.0 | -1.0 | +0.66| +0.33| +0.33| None (always OFF)
BCL-XL     | 0    | +1.0 | 0    | 0    | 0    | 0    | None
BAX(suppr) | 0    | -0.66| 0    | 0    | 0    | 0    | Inverse: active BAX=pro-death
CDKN2A(del)| +0.66| 0    | -1.0 | 0    | 0    | 0    | None (always OFF)
SMAD4(del) | +0.33| 0    | -0.66| +0.66| ◆    | +0.33| PSS: tumor still secretes TGFβ
RB1        | +0.33| 0    | -0.66| 0    | 0    | 0    | Downstream integrator
ZEB1       | -0.33| +0.33| 0    | +1.0 | 0    | +0.66| Go-vs-grow tradeoff with MYC
CDH1(repr) | 0    | 0    | 0    | +1.0 | 0    | 0    | Effect is when CDH1 is OFF
```

```
MMP2      | 0   | 0   | 0   | +1.0 | -0.33| 0   | Cycle 2: TGFβ liberation
TGFB1     | ◆   | 0   | ◆   | +0.66| +1.0 | +1.0 | GD/CCB depend on SMAD4
SHH       | 0   | 0   | 0   | 0   | +1.0 | +0.33| Paracrine only
HIF1A     | ◆   | +0.66| 0   | +0.66| +0.66| +0.66| GD: time-dependent
NRF2      | +0.33| +0.66| 0   | 0   | 0   | 0   | Primary output: ABCB1
ABCB1     | 0   | +1.0†| 0   | 0   | 0   | 0   | †DRUG-CONDITIONAL only

STROMAL CELLS:
Gene      | GD  | DR  | CCB | MI  | PSS | IET
-----------|------|------|------|------|------|------
ACTA2     | +0.33| +0.33| 0   | +0.66| +1.0 | +0.33
GLI1      | +0.66| +0.33| 0   | 0   | +0.66| 0
HAS2      | 0   | 0   | 0   | 0   | +1.0 | +0.33
COL1A1    | 0   | 0   | 0   | 0   | +1.0 | +0.33
TGFB1     | (as CAF feedback) | --  | +1.0 | --
HIF1A     | ◆   | +0.66| 0   | 0   | +0.66| +0.66
MMP2      | 0   | 0   | 0   | +0.33| -0.33| 0

◆ = requires conditional logic (see Phase G prompts)
† = drug-conditional (see Prompt B.3)
[S] genes (ACTA2, GLI1, HAS2, COL1A1) must NEVER activate in tumor cells (see Prompt B.6)
```

Appendix E: Execution Order Summary

```
PHASE A: Effect matrix data structure → compile-time truth table
PHASE B: Refined Boolean network →
  B.1: SMAD4 asymmetry (CRITICAL, do first)
  B.2: Go-vs-grow tradeoff
  B.3: Drug-conditional ABCB1
  B.4: Multi-axis KRAS
  B.5: MYC conditional death resistance
  B.6: Cell type scope enforcement
```

PHASE C: Confidence-weighted parameterization
PHASE D: BCL-XL toxicity in fitness function
PHASE E: Clinical trial failure validation (THE key acceptance test)
PHASE F: Stroma-as-restraint mechanics (required for Phase E to pass)
PHASE G: Context-dependent logic audit

DEPENDENCY CHAIN:
A → B.1 (matrix needed for B.1 to reference)
B.1 → E.1 (SMAD4 asymmetry needed for vismodegib test)
B.3 → E.1 (drug-conditional ABCB1 needed for resistance scenarios)
F.1 → E.1 (stroma restraint needed for vismodegib test to pass)
B.6 should be done EARLY (prevents hard-to-debug scope leaks)

COMPILE AND TEST AFTER EACH PHASE.
Phase E is the acceptance gate: if clinical validations pass, the behavior map is correctly implemented.